

Do LLM Judges Penalise the Script? A Script-Controlled Audit of LLM-as-Judge Scoring for Hindi in Devanagari versus Romanized Orthographies

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Abstract

Large language models now grade other language models, feeding leaderboards, reward models, and quality gates that all assume the writing system of an answer does not move its score. We test that assumption for Hindi, written natively in Devanagari and, by hundreds of millions online, in the Latin alphabet. We hold content fixed and vary only the script: 150 instruction-response pairs at three quality tiers, each in Devanagari and three romanizations (scholarly IAST, diacritic-stripped ASCII, user-style Hinglish), scored blind by seven judges from four families. We pre-specified the prediction that romanized text would score lower; the data reversed it. First, Gemini 3.6 Flash scores identical romanized content 2.2 to 3.4 points higher than Devanagari on a 100-point scale (all $p \leq 0.003$), concentrating at 6.5 to 7.8 points on medium-quality responses, where a judge has the most discretion; Gemini 3.1 Pro shows the same sign at one-third to one-half the size. Second, Qwen2.5-7B inflates scholarly IAST by 7.6 and ASCII by 6.0 points, reaching **+17.8** on low-quality items yet flat on Hinglish. Third, Claude judges (measured through a separate interface; §5) move opposite, the direction we pre-registered: one item per call, Sonnet 5 penalises by 2.7 to 12.3 points, Opus 5 by up to 3.4, Fable 5 near-invariant, GPT-5.6 near-robust, a suggestive capability ordering. Fourth, protocol decides what you see: with batch size the only varied factor, batching erases Sonnet’s -12.3 entirely ($\rightarrow +1.1$), while pairwise judging reverses Flash’s pointwise inflation into near-total Devanagari preference (zero ASCII wins in 600 trials across two judges, $p < 10^{-86}$). Fifth, prompting does not fix it: mitigation is at best partial on Gemini, flips sign under a rubric variant, and backfires on Qwen (up to $+11.8$ from a $+7.6$ baseline). A per-dimension decomposition shows Sonnet’s penalty is 68 percent a clarity judgment but leaks significantly into correctness, which identical content cannot differ on. Mechanism probes rule out token length ($|r| \leq 0.18$ with score shift) while rationales name the orthography on 41 to 57 percent of romanized items. Script bias in LLM judges is real, sign-opposite across families, protocol-sensitive in magnitude and sign, and not reliably promptable away.

1 Introduction

Evaluation of language models has largely been handed to other language models. An LLM judge reads a question and an answer and returns a score (Zheng et al., 2023) that then ranks systems on leaderboards, trains the next model as a reward signal, and gates what production systems ship. The reliability literature has catalogued many failure modes of these judges, including position bias (Wang et al., 2024), verbosity and self-enhancement bias (Zheng et al., 2023), self-preference driven by self-recognition (Panickssery et al., 2024), and taxonomies now spanning a dozen or more bias types (Ye et al., 2025; Zhou et al., 2026b). Every catalogue so far shares one silent assumption: that the writing system of the judged text does not move the score.

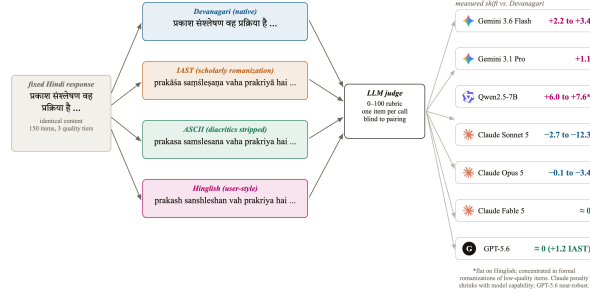


Figure 1: Study design. One Hindi response, four orthographies, seven blind judges. Any score gap is script bias by construction; measured outcomes previewed at right (§4).

That assumption matters most for languages with two written lives. Hindi is the clearest case. Its official script is Devanagari, but a very large share of Hindi on phones, in chat, and in social media is typed in the Latin alphabet, a practice called romanized Hindi or Hinglish (Sheth et al., 2025). The two forms carry identical content. A judge that scores them differently is not measuring quality; it is measuring orthography.

This paper asks one narrow question with a clean design: if we hold the language and the content of an answer fixed and change only its script, does an LLM judge’s score move? Because the content on both paths is identical (Figure 1), nothing else can explain a gap: not answer quality, not language difficulty, not topic. Prior multilingual judge studies could not make this claim because they compared different languages, so language and script were entangled (Hada et al., 2024b; Fu and Liu, 2025; Zhou et al., 2026a). We untangle them.

We recorded the predicted direction in the project log before collecting data: by analogy to the task-performance literature, where romanized input degrades accuracy by up to 24 points (Khullar et al., 2025) and inflates perplexity over 300-fold (Rajapakse and Weerasinghe, 2026), we expected romanized text to score lower. The data reversed the prediction: romanized Hindi scores higher on Gemini and Qwen judges, the inflation concentrates where a judge has discretion, and it persists or worsens when the judge is told to ignore the script. Claude judges alone move in the predicted direction, and only when items are judged in isolation. We report the reversal as found.

Our contributions:

- Script-controlled judge audit (§3): the first audit isolating script as a treatment: 150 fixed Hindi items, four orthographies, seven judges, four families, 17,100 scripted judgments (4,200 core, 2,400 protocol arms, 7,200 mitigation-battery, 1,200 sub-dimension, 600 pairwise, 600 cross-script, 600 test-retest, 300 anchored).
- A new, sign-opposite judge-bias axis (§4): up to +7.8 points of romanization inflation under Gemini 3.6 Flash and +17.8 under Qwen2.5-7B, against a penalty of up to −12.3 under Claude Sonnet 5 that shrinks with capability; orthography appears in no existing judge-bias taxonomy (Ye et al., 2025; Zhou et al., 2026b).
- A protocol effect that masks, and can even reverse, the bias (§4.3): a single-variable control shows batching alone erases Sonnet’s −12.3 ASCII penalty, and pairwise judging flips Flash’s pointwise inflation into near-total Devanagari preference (288/300 trials, zero ASCII wins), a warning for judge-bias audits and arena-style leaderboards alike.
- A six-arm mitigation battery that fails in three different ways (§4.4): prompting reduces the bias at best partially, flips its sign at worst, and amplifies it on the open-weights judge; the practitioner remedy is judge selection and script-stratified reporting, not prompting.
- Mechanism probes (§4.5): token length is ruled out as the driver, and the judges’ rationales show the script is perceived while the miscalibration happens anyway.

2 Related Work

LLM-as-judge and its biases. [Zheng et al. \(2023\)](#) established the paradigm and catalogued position, verbosity, and self-enhancement biases; [Wang et al. \(2024\)](#) flipped pairwise judgments by candidate order alone; [Panickssery et al. \(2024\)](#) traced self-preference to self-recognition; CALM ([Ye et al., 2025](#)) and JudgeBiasBench ([Zhou et al., 2026b](#)) quantify a dozen-plus bias types. Orthography is absent from all of these.

Multilingual judging. [Hada et al. \(2024b\)](#) calibrated GPT-4 judging against native speakers across eight languages, finding systematic overscoring, worst for low-resource languages. METAL ([Hada et al., 2024a](#)) and PARIKSHA ([Watts et al., 2024](#)) measured judge-human agreement across languages; MM-Eval ([Son et al., 2024](#)) and [Fu and Liu \(2025\)](#) stress-tested judges across up to 122 languages; BabelJudge ([KC, 2026](#)) measures Hindi reliability in native script only (see also [Zubiaga et al., 2026](#)). Closest to us, [Zhou et al. \(2026a\)](#) audit 23 languages, find lower-resource ones scored more generously, and note a Latin-script advantage observationally. In every one of these, script co-varies with language; none isolates it.

Script and romanization in LLMs. RomanSetu ([Husain et al., 2024](#)) showed romanization can serve as an efficient adaptation interface for Indic languages, building on a longer transliteration-adaptation history ([Purkayastha et al., 2023](#); [Xhelili et al., 2024](#); [Madhani et al., 2023](#)). On deployment, Script Gap ([Khullar et al., 2025](#)) measured task-accuracy drops up to 24 points on romanized inputs across five Indian languages, [Rajapakse and Weerasinghe \(2026\)](#) measured over 300-fold perplexity degradation on romanized Sinhala, and Indi-RomCoM ([Das et al., 2026](#)) benchmarked romanized code-mixed degradation (resources: [Sheth et al., 2025](#); [Yang and Chai, 2025](#)). Mechanistically, RomanLens ([Saji et al., 2025](#)) finds latent romanization inside multilingual models, and [Karne \(2026\)](#) shows concept representations are not script-invariant. All of this measures the model as a task performer or probes its internals; none measures it as a judge.

Tokenizer inequity. Tokenizers price languages unequally ([Petrov et al., 2023](#); [Ahia et al., 2023](#); [Ali et al., 2024](#); [Liang et al., 2023](#)); Indic byte-fallback fragmentation has been quantified as an eight-fold token tax ([Srivastava, 2026](#)), with morphology-aware remedies proposed ([Limisiewicz et al., 2024](#); [Brahma et al., 2025](#)). This literature stops at token counts and task accuracy; we test, and reject, its natural prediction for judge scores.

Indic evaluation. Indic benchmarks evaluate in the conventional native script only ([Ahuja et al., 2023](#); [Doddapaneni et al., 2023](#); [Singh et al., 2024](#); [Verma et al., 2025](#)); Airavata ([Gala et al., 2024](#)) exemplifies current practice, its Hindi generations GPT-4-judged in Devanagari with the judge unvalidated. Our result says that practice is sensitive to an unexamined variable.

3 Method

Dataset. We construct 150 Hindi instruction and response pairs (authored by a Claude-family model under human-specified tier definitions; §5 discusses the resulting self-preference exposure) across ten everyday domains, at three intended quality tiers of 50 items each: high (correct, complete, 80 to 140 words), medium (correct but visibly incomplete, 40 to 80 words), and low (curt, vague, or shallow, 15 to 40 words, a few with deliberate minor factual errors). Responses were authored in Devanagari and frozen before any judging. Tier intent was strongly realised: Devanagari means of 99.1, 51.4, and 15.2 under Gemini 3.6 Flash.

Conditions. Each frozen response is rendered in four orthographies. deva is the original Devanagari. iast is scholarly romanization with diacritics, produced deterministically with the indic-transliteration library ([Madhani et al., 2023](#)). ascii strips the diacritics from IAST, a proxy for plain-keyboard romanization. hinglish is a natural user-style respelling (for example kaise hain aap), produced under a strict no-paraphrase rule: every word, the word order, and all punctuation preserved. In every condition the instruction is rendered in the same orthography as the response (the ascii cell uses the IAST instruction), so a romanized response never answers a Devanagari question; §4.6 also measures the deliberately

Table 1: Mean within-item score shift (romanized minus Devanagari), bootstrap 95% CIs, $n=150$. Bold: permutation $p < 0.01$. All judges scored one item per call (§4.3).

Judge	IAST	ASCII	Hinglish
Gemini 3.6 Flash	+3.43 [+2.4, +4.6]	+2.17 [+0.8, +3.7]	+3.13 [+2.2, +4.1]
Gemini 3.1 Pro	+1.17 [+0.6, +1.8]	+1.10 [+0.4, +1.8]	+1.13 [+0.5, +1.8]
Qwen2.5-7B	+7.63 [+4.9, +10.4]	+6.03 [+3.3, +8.7]	+0.07 [-1.8, +1.9]
Claude Sonnet 5	-2.70 [-4.7, -0.8]	-12.29 [-15.6, -9.2]	-2.71 [-4.0, -1.4]
Claude Opus 5	-0.13 [-0.8, +0.6]	-3.43 [-4.4, -2.5]	-1.18 [-1.9, -0.5]
Claude Fable 5	+1.43 [+1.0, +1.9]	-0.30 [-1.1, +0.4]	+0.13 [-0.3, +0.6]
GPT-5.6	+1.21 [+0.2, +2.2]	+0.51 [-0.6, +1.7]	+0.62 [-0.2, +1.5]

mismatched cells. A native-speaker audit of a stratified 20-item sample of the hinglish condition confirmed 20/20 preserve content exactly; the audited sample ships with the released data.

Judges and protocol. Seven judges from four families score every item in every condition on a 0 to 100 rubric (correctness, completeness, helpfulness, clarity). Gemini 3.6 Flash and Gemini 3.1 Pro Preview are called through the API at temperature 0, one item per call, in randomized order, so the judge never sees two versions of the same item in one context. Claude Sonnet 5, Opus 5, and Fable 5 judge through blind agent sessions arranged in a Latin square: each session sees each item exactly once, in exactly one condition, with conditions rotated across four sessions. Qwen2.5-7B-Instruct runs open-weights under vLLM on one A10G GPU at temperature 0, one item per call, with top-20 logprobs recorded at the score position. GPT-5.6 is called one item per call at temperature 0 through a unified OpenAI-compatible gateway. No judge is told the study concerns scripts. The judging prompt asks for a score and a one-sentence reason; the reasons become data for the perception analysis (§4.5).

Follow-up batteries. Eight targeted batteries close the main alternative explanations (§4.6): a protocol control varying only batch size on a fixed interface; a per-dimension rubric decomposition on the main arms; an order-counterbalanced pairwise preference test; a cross-script control measuring the deliberately mismatched instruction/response cells; test-retest baselines for a deterministic and a stochastic judge; an anchored-pointwise variant with an in-context reference; and Benjamini-Hochberg FDR correction plus a logit-scale tier reanalysis. Verbatim prompts and full tables are in the appendix.

Analysis. The unit of analysis is the within-item paired difference, romanized minus Devanagari, per judge and condition. We report mean shifts with bootstrap 95 percent confidence intervals (10,000 resamples), two-sided sign-flip permutation tests (20,000 permutations), and the paired effect size d_z . All numbers in this paper are produced by one scripted analysis reading the raw score files; the analysis output ships with the paper.

Pre-registration. Before data collection we recorded the predicted direction: romanized scores lower than Devanagari. The observed direction is the opposite. We report all pre-registered analyses regardless.

4 Results

4.1 Three families, three bias profiles

Table 1 and Figure 2 are the study in one view. Gemini 3.6 Flash scores the same content higher when romanized: +3.43 for IAST ($p = 5 \times 10^{-5}$, $d_z = 0.49$), +2.17 for ASCII ($p = 0.003$), +3.13 for Hinglish ($p = 5 \times 10^{-5}$, $d_z = 0.52$); Gemini 3.1 Pro shows the same sign at one-third to one-half the size (all $p \leq 0.003$).

The open-weights judge changes shape. Qwen2.5-7B inflates formal romanizations dramatically, +7.63 for IAST and +6.03 for ASCII, but is indistinguishable from zero on natural Hinglish. Its inflation also lives in a different place: low-quality items shift +17.8 (IAST) and +17.4 (ASCII) points, meaning the small judge fails to recognise bad content once it

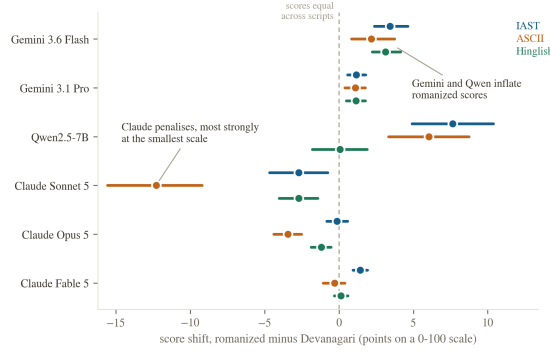


Figure 2: Score shift by judge and romanization, 95% bootstrap CIs, $n = 150$. Right of the dashed zero line means romanized scored higher.

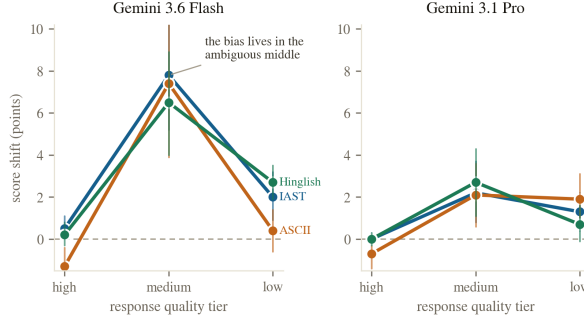


Figure 3: Shift by quality tier, Gemini judges, 95% bootstrap intervals, $n = 50$ per tier. High/low tiers sit near the score ceiling/floor.

is dressed in diacritics, while Hinglish, plentiful in web training data (Sheth et al., 2025), is judged as soberly as Devanagari.

The Claude family moves in the opposite direction, the one we pre-registered. Judged one item per call, Sonnet 5 penalises every romanization: -2.70 for IAST ($p = 0.007$), -12.29 for ASCII ($p = 5 \times 10^{-5}$, $d_z = -0.62$), and -2.71 for Hinglish ($p = 10^{-4}$). Opus 5 shows the same sign at a fraction of the size (-3.43 on ASCII, -1.18 on Hinglish), and Fable 5 is near-invariant, with one small inflation on IAST ($+1.43$). Within one family, then, the script penalty shrinks monotonically with model capability, and the pattern extends across families: GPT-5.6 is likewise near-robust (largest shift $+1.21$ on IAST, $p = 0.02$; ASCII and Hinglish indistinguishable from zero), though it still inflates low-tier romanized items by $+3.1$ to $+3.8$. Sonnet’s rationales say why out loud: they call romanized responses “awkward IAST-transliterated Hindi rather than natural Devanagari,” treating the orthography itself as a quality defect. Because our rubric asks for a clarity sub-score, and stripped ASCII is plausibly less clear to a competent Hindi reader, part of the Sonnet penalty could be a legitimate clarity judgment; a per-dimension decomposition (§4.6) confirms clarity carries 68 percent of it but shows a significant residue on correctness and helpfulness, dimensions identical content cannot differ on.

Three conclusions follow. Script bias in LLM judges exists, in both directions: the same romanized answer is overpaid by one family and taxed by another. It is a property of the judge, not the task: same items, same protocol, three behaviours across four families, with a capability gradient. And it is not monotone in naturalness: Gemini inflates every romanization, Qwen inflates the formal romanizations it saw least in training, and Claude taxes hardest the stripped-ASCII form that most visibly degrades the text.

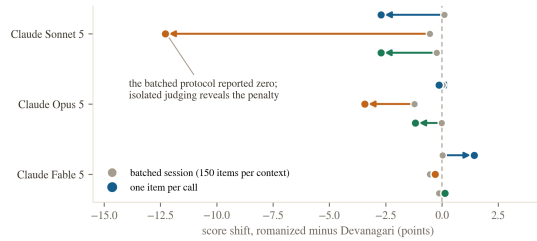


Figure 4: Two protocols, two answers. Batched sessions (grey) versus one item per call (colored) for the same three Claude judges; batching reports near-zero, isolation reveals penalties up to -12.3 .

4.2 The bias lives where the judge has discretion

Figure 3 splits the shift by quality tier. Under Flash, high-quality items shift at most 1.3 points against a ceiling near 99, low-quality at most 2.7 against a floor near 15, while medium-quality items, near 51 in Devanagari, inflate by $+7.8$ (IAST), $+7.4$ (ASCII), and $+6.5$ (Hinglish), all $p \leq 1.5 \times 10^{-4}$. The same shape holds for Pro at smaller magnitude. This is the pattern one expects if romanization blurs the judge’s view of quality: content whose flaws are obvious stays low, content whose merits are obvious stays high, and ambiguous content drifts toward a generous middle. It is also the pattern a purely mechanical ceiling-and-floor compression would produce with no discretion story at all; a logit-scale reanalysis (§4.6), which removes exactly that compression, still finds the medium tier’s shift largest, so the discretion reading survives the transform. Appendix Figure 7 shows one such drift verbatim: one refrigerator explanation, four orthographies, a thirty-point spread under one judge.

4.3 The measurement protocol can hide the bias

Our first Claude measurements used batched agent sessions, 150 items per context with conditions rotated in a Latin square; all three Claude judges appeared script-invariant, shifting at most 0.54 points in eight of nine comparisons. Rerunning the identical items one per call flipped the answer (Figure 4): the invariance was a batch artifact. Because those two arms also differed in interface and decoding temperature, we ran a single-variable control: the same Sonnet 5, the same headless interface, the same per-item rubric and sampling, with items grouped 25 per call. Batch size alone reproduces the masking: the -12.29 ASCII penalty becomes $+1.12$, and the high-tier penalty collapses from -24.3 to -3.6 (full table in the appendix). A judge that has just scored dozens of items appears to anchor its scale on content quality and stops reacting to surface form; shown one item in isolation, it lets the orthography move the score. Batched audits, common because they are cheap, can therefore mask a twelve-point bias that one-item-per-call production judging would feel in full. Judge-bias measurements should state their context protocol.

4.4 Prompting does not reliably fix it

The remedy every practitioner would try first is a line in the judge prompt. We tested six: a plain instruction, an explicit warning naming both scripts, a fairness framing, a transliterate-first instruction, a script-blind persona, and a decomposed rubric forcing four sub-scores summing to 100; each ran over the full four-condition protocol on two judge families (Figure 5).

On Gemini 3.6 Flash, no arm closes the gap. The best, transliterate-first, cuts medium-tier inflation from $+7.8$ to $+4.4$; the persona arm does nothing ($+7.7$); and the decomposed rubric overshoots, flipping ASCII from $+7.4$ to -2.8 , trading inflation for penalty rather than removing bias. On Qwen2.5-7B the battery backfires uniformly: every instruction increases IAST inflation above the $+7.6$ no-instruction baseline, up to $+11.8$ under the fairness framing, and the script-blind persona even induces a $+4.6$ Hinglish inflation where the baseline judge was flat. Mentioning scripts in the prompt appears to direct the smaller

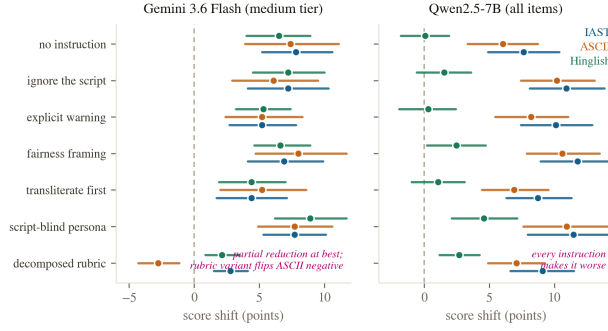


Figure 5: Six mitigation arms, two families, no reliable fix. Left: Gemini 3.6 Flash, medium tier, $n = 50$. Right: Qwen2.5-7B, all items, $n = 150$: every arm increases inflation.

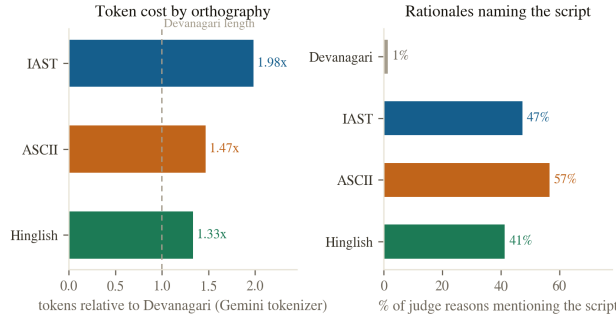


Figure 6: Left: token cost relative to Devanagari, Gemini tokenizer, $n = 150$. Right: share of judge rationales naming the script, Gemini 3.6 Flash, $n = 150$ per condition.

judge’s attention to the very feature it should ignore. Prompt-level mitigation is therefore unreliable in the strongest sense: its effect ranges from partial reduction through sign reversal to amplification, consistent with debiasing results showing prompt-level fixes underperform training-level ones (Zhou et al., 2026b). The practitioner remedy remains judge selection and script-stratified reporting, not prompting. The mitigation battery tests pointwise absolute scoring; whether these instructions transfer to pairwise preference judging is untested, and §4.6 shows the pairwise protocol itself behaves very differently.

4.5 Mechanism: not token length; the script is perceived

Two probes narrow the explanation space. First, tokenization: the tokenizer-fairness literature predicts length disparities drive downstream behaviour (Petrov et al., 2023; Ahia et al., 2023; Srivastava, 2026). Measured with the Gemini tokenizer, romanized Hindi is the expensive form, costing 1.98 (IAST), 1.47 (ASCII), and 1.33 (Hinglish) times the Devanagari tokens (Figure 6, left), itself a finding: it reverses the romanization-efficiency premise of 2 to 4-fold token savings (Husain et al., 2024); modern tokenizers have closed the gap (Brahma et al., 2025). But token inflation does not explain the score inflation: the per-item correlation between token ratio and score shift is $r = +0.11$ (IAST), $+0.05$ (ASCII), and -0.08 (Hinglish), and no tier-restricted correlation exceeds $|r| = 0.18$. Longer input is not what earns the higher score. The top-20 logprobs recorded at the Qwen score position offer a partial distributional view: because they were captured at the first generated token only, they describe the leading digit of the score, not the full 0-100 value. On that limited view, romanized conditions shift the leading-token expectation upward (IAST 6.33 versus Devanagari 5.70) with essentially unchanged within-call spread, consistent with a mode move rather than a widening; a full-sequence distributional analysis remains future work.

Second, perception. The judge’s rationales show it sees what it is scoring (Figure 6, right): under Flash, 47 percent of IAST, 57 percent of ASCII, and 41 percent of Hinglish rationales

explicitly mention the script, against 1 percent for Devanagari; some name the orthography as a deficiency while scoring the item above its Devanagari twin. The bias operates despite awareness, echoing behaviourally the finding that internal representations are not script-invariant (Karne, 2026; Saji et al., 2025).

4.6 Follow-up batteries: closing the confounds

Eight targeted batteries (4,500 further judgments: 600 GPT core, 600 batched control, 1,200 sub-dimension, 600 pairwise, 600 cross-script, 600 test-retest, 300 anchored; tables in the appendix) address the main alternative explanations.

Noise floor. Thirty stratified items, deva and ascii, five identical calls each: mean within-item standard deviation is 1.66 points for Flash at temperature 0 and 2.00 for Sonnet under the headless interface’s default sampling. Sonnet’s -12.3 , Qwen’s $+6.0$ to $+7.6$, and the medium-tier inflations sit five to seven of these standard deviations out; shifts near ± 1 do not.

Is the penalty just clarity? Under the decomposed four-dimension rubric, Sonnet’s ASCII shift totals -7.14 : clarity carries -4.82 (68 percent), but correctness drops -1.57 ($p = 5 \times 10^{-5}$) and helpfulness -0.54 ($p = 0.001$), dimensions on which identical content cannot legitimately differ. Flash’s shift is pure clarity (-2.29 ; the other three near zero). The answer is judge-dependent, and the non-clarity residue is bias by construction.

Is it question-answer script mismatch? No, but the mismatch effect is real: an ASCII response answering a Devanagari instruction scores 13.5 (Flash) and 16.1 (Sonnet) points below the same response answering a script-matched instruction (both $p = 5 \times 10^{-5}$). Mismatch is a separate, larger penalty, and every headline number is free of it by construction (§3).

Does pairwise judging agree? No, starkly. Shown the deva and ascii twins of the same item in both orders, Flash prefers the Devanagari version in 288 of 300 trials and Sonnet in 293 of 300, with zero ASCII wins for either (sign-test $p < 10^{-86}$). Flash inflates ASCII by $+2.2$ pointwise yet prefers Devanagari 96 percent of the time pairwise: the two standard protocols disagree not in degree but in sign, and preference-trained reward models sit on the pairwise side. An anchored-pointwise variant (the frozen Devanagari original as an in-context reference) points the same way: the ASCII inflation vanishes (-0.97 , $p = 0.65$) and reverses to -10.4 on high-quality items, suggesting the pairwise verdict is the anchored assessment and unanchored pointwise is the outlier; the deva cell’s response equals its reference, so this is directional evidence only.

Multiple comparisons and scale artifacts. Benjamini-Hochberg correction over all 198 reported p-values leaves 108 significant at $q = 0.05$ (effective threshold $p \leq 0.0249$), including every headline effect. A logit-transform reanalysis of the tier pattern, which removes ceiling-and-floor compression, still places the largest shift at the medium tier for all three romanizations, so the discretion account survives (§4).

5 Limitations, narrated

We narrate the study’s weaknesses as part of the story; several were speed-and-budget design choices a reader should weigh.

There is no human reference score anywhere in this study. Every headline number is a shift relative to Devanagari, an anchor choice rather than validated ground truth: the data equally support judges under-scoring Devanagari, and words like “inflates,” “penalises,” and “miscalibration” assert a direction of error the design cannot establish. Read them as shorthand for “shifts relative to Devanagari” pending a human-anchored replication. The per-dimension decomposition (§4.6) bounds but does not eliminate the related clarity concern: it shows the non-clarity residue of Sonnet’s penalty is bias by construction, but whether the clarity component itself is calibrated to what a human reader experiences still requires human ratings.

The dataset is model-authored: all 150 items and their tiers were generated by a language model, then frozen; model-authored Hindi may differ from human Hindi in ways that interact with romanization. The hinglish condition is a model respelling under a strict no-paraphrase rule; a native-speaker audit of a stratified 20-item sample confirmed 20/20 preserve content, word order, and punctuation exactly, though the audit covers only hinglish, not IAST or ASCII, and does not certify the content as representative of human Hindi.

The Claude judges were measured under two protocols: batched agent sessions and headless one-item-per-call calls to the pinned model ids. The headline numbers use the isolated protocol, matching Gemini and Qwen; its one residual difference is that the headless interface exposes no temperature control, so Claude uses default sampling where the API arms use temperature 0. The single-variable batching control (§4.6) now attributes the masking effect to batch size alone, but the cross-family comparison, the Gemini-versus-Sonnet sign reversal, still varies judge family, interface, and decoding regime at once. The 150 items were authored by a Claude-family model, so the frozen Devanagari originals are that family’s own emissions and the romanized conditions are transforms of them. A self-recognition account (Panickssery et al., 2024) would predict exactly the Claude judges’ preference for the authored surface form; blind single-item calls do not exclude it. Read the Claude-direction results with that exposure in mind. The Gemini, Qwen, and GPT results are unaffected, and a control set authored by a different family is the clean fix not yet run.

The core arms ran one seed per item per condition; the test-retest batteries (§4.6) now provide noise yardsticks for both a deterministic and a stochastic judge, and the headline effects clear them several-fold, but per-item shifts near a point remain indistinguishable from sampling noise. The pairwise battery demonstrates a preference-level consequence, but we still do not show a ranking flip between closely matched real systems; our tiers are separated by design, so ordering stability among them is unsurprising. FDR correction is now reported (§4.6); individual small cells that do not survive it (90 of 198 tests) should be read as unconfirmed.

Scale and scope are modest: one language, 150 items, seven judges, six prompt-level mitigation arms on two families, no training-level intervention. The quality tiers are coarse; the logit reanalysis (§4.6) defends the discretion account against scale compression, but a continuous-quality design would test it more sharply. The capability ordering rests on four checkpoints across two families with no independent capability metric: suggestive, not a scaling law. Extending to other digraphic settings (Karne, 2026; Rajapakse and Weerasinghe, 2026) is the natural next step.

Finally, the pre-registration was directional and informal (a project-log entry, not a registry filing); we state it because it disciplined our reporting of a reversed prediction.

6 Conclusion

We held Hindi content fixed, changed only its writing system, and asked seven LLM judges to grade it. The writing system moved the grades, in both directions: Gemini judges inflate romanized Hindi by 2 to 3 points overall and 7 to 8 on ambiguous-quality content; a 7B open-weights judge inflates formal romanizations by up to 18 points on low-quality content; Claude judges tax romanization instead, up to 12.3 points at the smallest scale in a suggestive capability ordering, while GPT-5.6 is near-robust; batch size alone erases the largest penalty; the judge that inflates ASCII pointwise prefers Devanagari in 96 percent of pairwise trials; the penalty is majority clarity but leaks into correctness; six mitigations range from partial to counterproductive; token length explains none of it; and the rationales prove the script is seen. The bias axis is real, judge-dependent in size and sign, protocol-sensitive in magnitude and sign, and it belongs in every judge-bias taxonomy (Ye et al., 2025; Zhou et al., 2026b) and multilingual evaluation checklist (Hada et al., 2024b; Son et al., 2024), though establishing it as judge error rather than non-invariance still requires a human anchor (§5). Until judges are audited for script invariance, Hindi evaluations should report native-script and romanized slices separately, and should say which judge produced the numbers.

Reproducibility Statement

All datasets, per-call score files, the scripted analysis, figure build scripts, and spend logs accompany this submission and will be released publicly upon acceptance. Every number in this paper traces to a single analysis script reading the raw judgment files, described in §3. Total marginal compute cost of the study was under 6 US dollars of metered API spend plus roughly one GPU-hour on a single accelerator for the open-weights judge; Claude judging ran through a separate subscription-billed interface not captured in that figure.

LLM Usage Statement

Large language models were used as the subjects of this study (the seven judges under audit) and as engineering assistants for writing analysis code, drafting figures, and copy-editing this manuscript. All experimental design, statistical choices, interpretation of results, and claims are the authors'; every number reported was checked against the scripted analysis output described in §3 before being written into the text.

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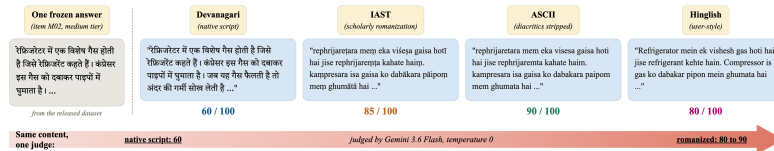


Figure 7: Qualitative example (item M02). Same content, four orthographies, one judge: scores of 60 (Devanagari), 85, 90, and 80 (romanized), verbatim from the released data.

Table 2: GPT-5.6 core battery (one item per call, temperature 0, unified OpenAI-compatible gateway), $n=150$ per condition (149 for ASCII; one unparsed call).

Condition	Shift	95% CI	p	by tier (high/med/low)
IAST	+1.21	[+0.18, +2.23]	0.023	+0.40 / -0.60 / +3.84
ASCII	+0.51	[-0.61, +1.66]	0.384	-1.84 / +0.26 / +3.06
Hinglish	+0.62	[-0.23, +1.48]	0.157	+0.02 / +0.56 / +1.28

Table 3: Per-dimension decomposition of the ASCII-versus-Devanagari shift (0-25 scale per dimension), $n=150$ pairs per judge. Sonnet 5 via the headless one-per-call interface; Flash via its native API at temperature 0.

Judge	Correctness	Completeness	Helpfulness	Clarity
Claude Sonnet 5	-1.57 $p=5 \times 10^{-5}$	-0.21 $p=0.18$	-0.54 $p=0.001$	-4.82 $p=5 \times 10^{-5}$
Gemini 3.6 Flash	-0.39	+0.37	+0.22	-2.29

FDR correction. Benjamini-Hochberg over all 198 reported p-values: 108 remain significant at $q=0.05$, including every effect named in the abstract. The full per-test table ships in `analysis_extra.json`.

Logit-scale tier reanalysis (Flash). Transforming scores by $\text{logit}(s/100)$ (clipped to $[0.5, 99.5]$) before differencing removes bounded-scale compression. The medium tier retains the largest shift under all three romanizations (logit shifts +0.29 to +0.35, versus at most +0.22 for high and low tiers, and -0.48 for high-tier ASCII), so the discretion account survives the transform.

Qwen leading-token distribution. Top-20 logprobs at the first score token (leading digit for two-digit scores): expectation 5.70 (deva), 6.33 (IAST), 6.17 (ASCII), 5.73 (Hinglish); within-call SD 0.60 to 0.62 throughout, consistent with a mode move rather than a widening.

C Per-tier shifts for every judge

Mean within-item shift versus Devanagari by quality tier (high / medium / low, $n=50$ per tier), one-item-per-call arms. Rows are the headline (isolated-protocol) numbers behind Table 1 and Figure 3.

D Mitigation battery, full per-arm table

Gemini 3.6 Flash, mean shift versus Devanagari per arm (all items $n=150$; medium tier $n=50$ in parentheses). Qwen per-arm numbers are in §4.4.

Table 4: Pairwise deva-versus-ascii preference, both presentation orders, 300 trials per judge.

Judge	Deva wins	ASCII wins	Ties	Sign-test p	Deva-first / ASCII-first splits
Gemini 3.6 Flash	288	0	12	4.0×10^{-87}	149-0-1 / 139-0-11
Claude Sonnet 5	293	0	7	1.3×10^{-88}	—

Table 5: Protocol control, Claude Sonnet 5: identical items, identical headless interface and per-item rubric; only batch size differs. $n=150$ per cell.

Protocol	IAST	ASCII	Hinglish	ASCII by tier (h/m/l)
One per call	-2.70	-12.29	-2.71	-24.28 / -12.96 / +0.38
Batched (25/call)	+2.64	+1.12	-1.29	-3.62 / +3.48 / +3.50

Table 6: Cross-script control: deliberately mismatched instruction/response cells versus the matched cells of the core arms. $n=150$ per cell, both $p = 5 \times 10^{-5}$ except where shown.

Mismatched cell vs matched baseline	Flash	Sonnet 5
Deva instruction + ASCII response, vs matched ASCII	-13.50 [-15.6, -11.5]	-16.08 [-19.6, -12.5]
IAST instruction + Deva response, vs matched Deva	+3.10 [+2.1, +4.2]	-3.71 [-5.5, -2.0]

Judge	Condition	High	Medium	Low
Gemini 3.6 Flash	IAST	+0.50	+7.80	+2.00
	ASCII	-1.30	+7.40	+0.40
	Hinglish	+0.20	+6.50	+2.70
Gemini 3.1 Pro	IAST	0.00	+2.20	+1.30
	ASCII	-0.70	+2.10	+1.90
	Hinglish	0.00	+2.70	+0.70
Qwen2.5-7B	IAST	-1.50	+6.60	+17.80
	ASCII	-1.90	+2.60	+17.40
	Hinglish	-0.20	-1.40	+1.80
Claude Sonnet 5	IAST	-6.74	-3.80	+2.44
	ASCII	-24.28	-12.96	+0.38
	Hinglish	-5.58	-3.90	+1.34
Claude Opus 5	IAST	-1.02	+0.12	+0.50
	ASCII	-6.18	-4.08	-0.02
	Hinglish	-1.82	-1.94	+0.22
Claude Fable 5	IAST	-0.28	+2.78	+1.80
	ASCII	-3.78	+1.46	+1.42
	Hinglish	-0.06	-0.32	+0.78
GPT-5.6	IAST	+0.40	-0.60	+3.84
	ASCII	-1.84	+0.26	+3.06
	Hinglish	+0.02	+0.56	+1.28

Arm	IAST	ASCII	Hinglish
None (baseline)	+3.43 (+7.80)	+2.17 (+7.40)	+3.13 (+6.50)
P1 plain	+2.73 (+7.20)	+2.03 (+6.10)	+3.07 (+7.20)
P2 explicit warning	+2.33 (+5.20)	+1.73 (+5.20)	+2.23 (+5.30)
P3 fairness	+3.00 (+6.90)	+3.07 (+8.00)	+2.93 (+6.60)
P4 transliterate-first	+2.10 (+4.40)	+1.93 (+5.20)	+1.83 (+4.40)
P5 script-blind persona	+3.27 (+7.70)	+2.97 (+7.70)	+3.80 (+8.90)
R1 decomposed rubric	+1.47 (+2.78)	-2.36 (-2.76)	+1.37 (+2.14)